

1. What is software?

Software is a set of instructions for the hardware.

2. What is Python and what are its applications?

Python is an object-oriented programming language that is easy to learn and simple to implement.

Applications of Python

Python is a versatile language that has applications in almost every field

- Artificial intelligence (AI)
- Machine Learning (ML)
- Big Data
- Smart Devices/Internet of Things (IoT)
- Cyber Security
- Game Development
- Backend Development, etc.

3. What are the features of Python?

Features of Python:

- Easy to learn & code
- Open Source Programming Language
- Object-Oriented Language
- Dynamic Typed Language
- Large Standard Library

4. Is Python case-sensitive?

Yes, Python is case-sensitive. The **username**, **UserName**, and **userName** are three different variables, and using these names interchangeably causes an error.

Code

PYTHON

```
1 username = "Rahul"
2 print(username)
3 print(userName)
```

Output

Rahul

```
NameError: name 'userN
```

5. Is Python a dynamically typed programming language?

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Yes, Python is a dynamically typed language. This means that in Python the type checking of a variable is done only as code runs, and the type of a variable is allowed to change over its lifetime. There is no need to declare the type of the variable

While programming languages like C, Java, C++, etc are statically typed languages where we cannot change the data type of a variable during the execution of the program.

Code

PYTHON

```
1 x = 6
2 print(type(x))
3 x = 'Rahul'
4 print(type(x))
```

Output

```
<class 'int'>
<class 'str'>
```

6. What are the advantages of Python over Java?

Basis of Comparison	Python
Learning curve	Easy to learn
Typing	Dynamically-typed
Syntax	Easy to read and remember
Applications	Artificial Intelligence, Data Science and Machine Learning applications
Code Length	Fewer lines of code compared to Java
Example Program	print("Hello World")

7. How to perform the arithmetic operations using Python?

Addition

The addition is denoted by

+ sign. It gives the sum of two numbers.

Code

PYTHON

```
1 print(2 + 5)
2 print(1 + 1.5)
```

Output

```
7
2.5
```

Subtraction

The subtraction is denoted by

- sign. It gives the difference between the two numbers.

Code

PYTHON

```
1 print(5 - 2)
```

Output

3

Multiplication

The multiplication is denoted by

* sign.

Code

PYTHON

```
1 print(2 * 5)
2 print(5 * 0.5)
```

Output

10

2.5

Division

The division is denoted by

/ sign.

Code

PYTHON

```
1 print(5 / 2)
2 print(4 / 2)
```

Output

```
2.5
2.0
```

Modulus

To find the remainder between two numbers, we use the Modulus operator

%

Code

PYTHON

```
1 print(6 % 3)
```

Output

```
0
```

Exponent

To calculate **a power b**, we use Exponent Operator

**

Code

PYTHON

```
1 print(2 ** 3)
```

Output

8

8. What is floor division?

To find integral part of quotient we use Floor Division Operator

// .

- $a // b$

Code

PYTHON

```
1 print(3 // 2)
```

Output

1

9. What is Operator Precedence in Python?

The operator precedence determines which operator is executed first if there is more than one operator in an expression.

The operator precedence in Python is listed in the following table. It is in descending order (the upper group has higher precedence than the lower ones).

Operators

()

**

+X , -X , ~X

*, /, //, %

+, -

<<, >>

&

^

|

==, !=, >, >=, <, <=

not , in , not in

not

and

or



BODMAS

The standard order of evaluating an expression

- *Brackets* (B)
- *Orders* (O)
- *Division* (D)
- *Multiplication* (M)
- *Addition* (A)
- *Subtraction* (S)

Expression:

$$(5 * 2) + (3 * 4 + 4 / 2)$$

Step by Step Explanation

$$\begin{aligned} &(5 * 2) + (3 * 4 + 4 / 2) \\ &(10) + (3 * 4 + 2) \\ &(10) + (12 + 2) \\ &(10) + (14) \\ &24 \end{aligned}$$

Code

PYTHON

```
1 print((5 * 2) + (3 *
```

Output

24

10. What is a Variable?

Variables are like containers for storing values.

Assigning Value to Variable

The following is the syntax for assigning an integer value

10 to a variable age

PYTHON

```
1 age = 10
```

Here the equals to

= sign is called an **Assignment Operator** as it is used to assign values to variables.

11. What are Data Types?

In programming languages, every value or data has an associated type to it known as data type.

Some commonly used data types

- String
- Integer
- Float

- Boolean

This data type determines how the value or data can be used in the program. For example, mathematical operations can be done on Integer and Float types of data.

12. What are the numeric data types in Python?

The Numeric Data Types in Python are:

- Integers
- Float
- Complex Numbers

Code

PYTHON

```
1 a = 10
2 print("Type of a: ",
3
4 b = 10.0
5 print("Type of b: ",
6
7 c = 10 + 20j
8 print("Type of c: ",
```

Output

```
Type of a: <class 'int'>
Type of b: <class 'float'>
Type of c: <class 'complex'>
```

13. What is meant by mutability? Name some mutable data types?

Mutable means capable of being changed. In Python, objects whose value can be changed are said to be mutable.

Some of the mutable data types in Python are list, dictionary, set and user-defined classes.

14. What is meant by immutability? Name some immutable data types?

Immutable means capable of not being changed. In Python, objects whose value cannot be changed are said to be immutable.

Some of the immutable data types in Python are tuple, integer, boolean, string, etc.

15. What are the differences between the mutable and immutable data types?

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Mutable and **immutable** are terms primarily used to refer to whether an object can be changed (modified) after it's created.

Immutable Data Types

Cannot be changed after creation

Can be used as dictionary keys

Reduces potential bugs from unintended side effects

Examples: `int` , `float` , `str` , `tuple`

Creating a new instance doesn't affect the original object

For some operations, immutable data types may require the creation of many temporary objects which can be inefficient



16. What is type conversion or type casting?

Converting the value of one data type to another data type is called **Type Conversion** or **Type Casting**.

We can convert

- String to Integer
- Integer to Float
- Float to String and so on.

String to Integer

`int()` converts valid data of any type into integer

Code

PYTHON

```
1 a = "5"
2 a = int(a)
3 print(type(a))
4 print(a)
```

Output

```
<class 'int'>
5
```

Integer to String

`str()` converts data of any type into a string.

Code

PYTHON

```
1 a = input()
2 a = int(a)
```

```
3 a = input()  
4 b = int(b)  
5 result = a + b  
6 print("Sum: " + str(
```

Input

2
3

Output

Sum: 5

Similarly,

- `float()` -> Converts to a float data type
- `bool()` -> Converts to a boolean data type

17. What is a String?

A String is a stream of characters enclosed within quotes.

Stream of Characters

- Capital Letters (A – Z)
- Small Letters (a – z)
- Digits (0 – 9)

- Special Characters (~ ! @ # \$ % ^ . ? ,)
- Space

Some examples:

- "Hello, World!"
- "some@example.com"
- "1234"

18. What is String Slicing?

Obtaining a part of a string is called String Slicing.

Syntax:

```
variable_name[start_index
```

- `end_index` is not included in the slice.

Code

PYTHON

```
1 message = "Hi Ravi"
2 part = message[3:7]
3 print(part)
```

Output

Ravi

Slicing to End

If the end index is not specified, slicing stops at the end of the string.

Code

PYTHON

```
1 message = "Hi Ravi"  
2 part = message[3:]  
3 print(part)
```

Output

Ravi

Slicing from Start

If the start index is not specified, slicing starts from the index 0.

Code

PYTHON

```
1 message = "Hi Ravi"  
2 part = message[:2]  
3 print(part)
```

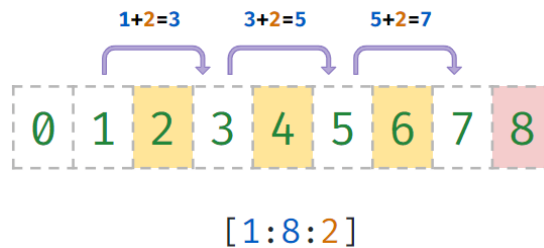
Output

Hi

Extended Slicing

Syntax:

`variable[start_index:end_index:step]`



Step determines the increment between each index for slicing.

Code

PYTHON

```
1 a = "Waterfall"
2 part = a[1:8:2]
3 print(part)
```

Output

aefl

19. How to reverse a string?

A string can be reversed using extended slicing.

Syntax:

```
variable[start:end:neg
```

-1 for step will reverse the order of the characters.

Code

PYTHON

```
1 string_1 = "Program"
2 string_2 = string_1[
3 print(string_2]
```

Output

margorP

20. What is string capitalize() in Python?

The

capitalize() method converts the first character of a string to an

uppercase letter and all other alphabets to lowercase.

Code

PYTHON

```
1 sentence = "proGraMm
2 capitalized_string =
3 print(capitalized_st
```

Output

Programming

21. What is string replace() in Python?

The

replace() returns a new string after replacing all the occurrences of the old substring with the new substring.

Syntax:

```
str_var.replace(old, new)
```

Code

PYTHON

```
1 sentence = "teh cat  
2 sentence = sentence.  
3 print(sentence)
```

Output

the cat and the dog

22. What is `round()` function?

Rounds the float value to the given number of decimal digits.

Syntax:

```
round(number, digits(optional))
```

`digits` -> defines the number of decimal digits to be considered for rounding.

- When `digits` not specified, the default value is **0**.

Code

PYTHON

```
1 a = round(3.14159, 2)
2 print(a)
3 a = round(5.6777)
4 print(a)
```

Output

```
3.14
6
```

23. How to write comments in Python?

A comment starts with a hash

#

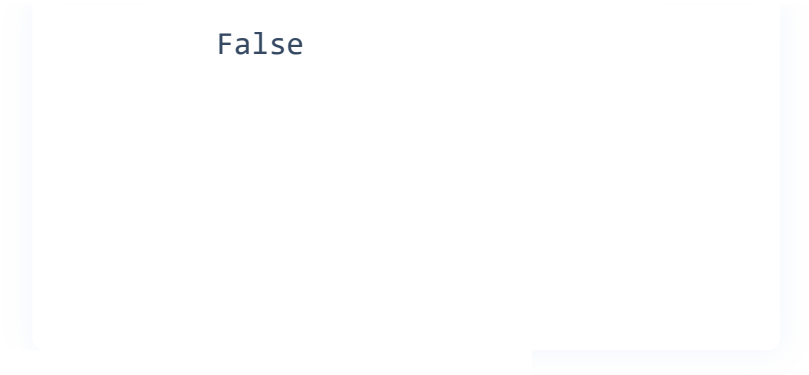
It can be written in its own line next to a statement of code.

Code

PYTHON

```
1 n = 5
2 # Finding if Even
3 even = (n % 2 == 0)
4 print(even) # prints
```

Output



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